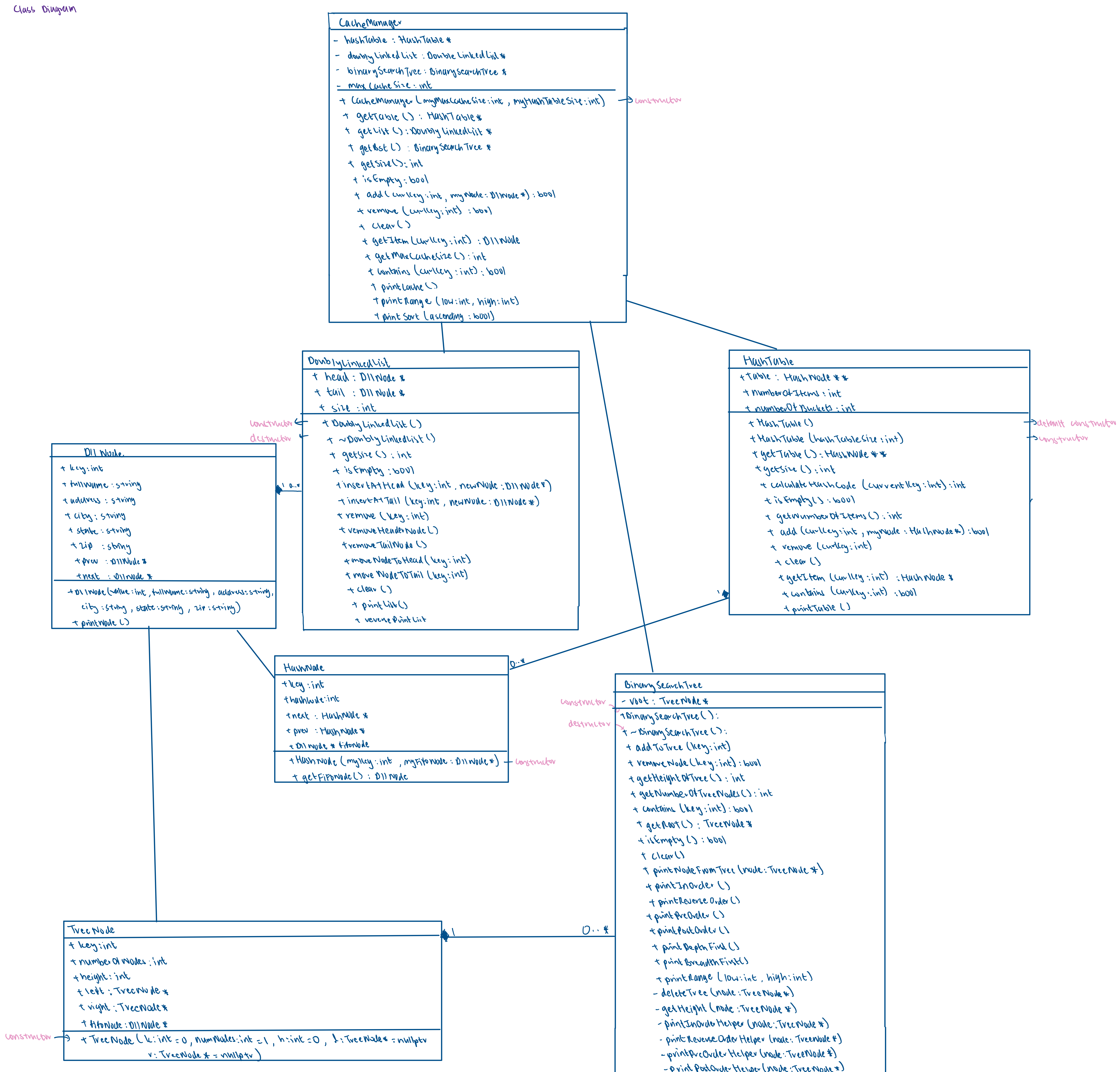
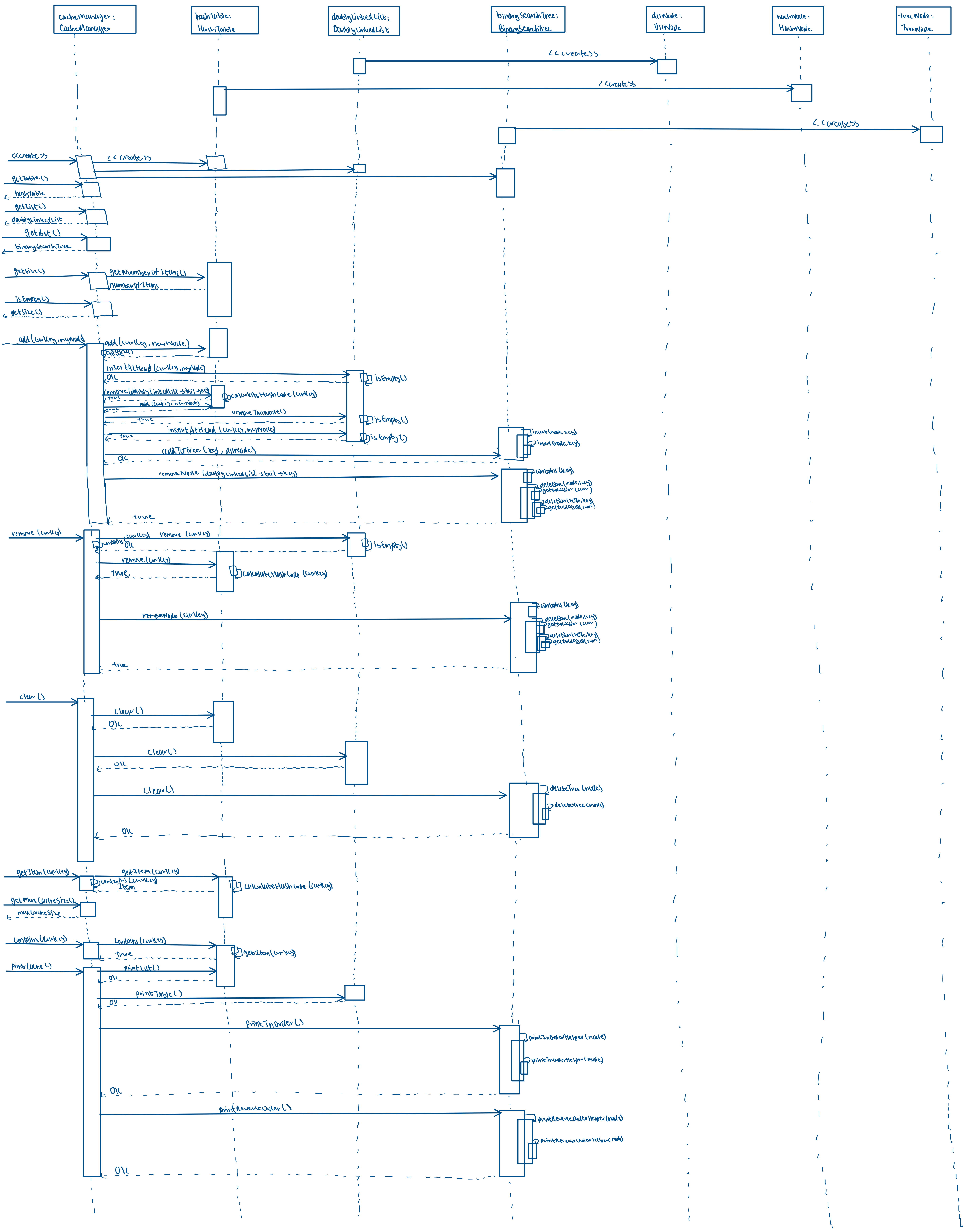
Milestone 5 UMLs

Tuesday, 28 April 2026 9:25 PM

Class Diagram


CacheManager
- hashTable : HashTable *
- doublyLinkedList : DoubleLinkedList *
- binarySearchTree : BinarySearchTree *
- maxCacheSize : int
+ CacheManager (myMaxCacheSize : int , myHashTableSize : int)
constructor
+ getTable () : HashTable *
+ getList () : DoublyLinkedList *
+ getBst () : BinarySearchTree *
+ getSize () : int
+ isEmpty : bool
+ add (currKey : int , myNode : DllNode *) : bool
+ remove (currKey : int) : bool
+ clear ()
+ getItem (currKey : int) : DllNode
+ getMaxCacheSize () : int
+ contains (currKey : int) : bool
+ printCache ()
+ printRange (low : int , high : int)
+ printSort (ascending : bool)
DoublyLinkedList
+ head : DllNode *
+ tail : Dll Node *
+ size : int
constructor
destructor
+ DoublyLinkedList ()
+ ~DoublyLinkedList ()
+ getSize () : int
+ isEmpty : bool
+ insertAtHead (key : int , newNode : DllNode *)
+ insertAtTail (key : int , newNode : DllNode *)
+ remove (key : int)
+ removeHeaderNode ()
+ removeTailNode ()
+ moveNodeToHead (key : int)
+ moveNodeToTail (key : int)
+ clear ()
+ printList ()
+ reversePrintList
HashTable
+ table : HashNode **
+ numberOfItems : int
+ numberOfBuckets : int
+ HashTable ()
default constructor
+ HashTable (hashTableSize : int)
constructor
+ getTable () : HashNode **
+ getSize () : int
+ calculateHashCode (currentKey : int) : int
+ isEmpty () : bool
+ getNumberOfItems () : int
+ add (currKey : int , myNode : HashNode *) : bool
+ remove (currKey : int)
+ clear ()
+ getItem (currKey : int) : HashNode *
+ contains (currKey : int) : bool
+ printTable ()
Dll Node
+ key : int
+ fullName : string
+ address : string
+ city : string
+ state : string
+ zip : string
+ prev : DllNode *
+ next : DllNode *
+ DllNode (value : int , fullName : string , address : string , city : string , state : string , zip : string)
+ printNode ()
1
0...*
HashNode
+ key : int
+ hashNode : int
+ next : HashNode *
+ prev : HashNode *
+ DllNode * fifoNode
+ HashNode (myKey : int , myFifoNode : DllNode *)
constructor
+ getFifoNode () : DllNode
0..*
1
BinarySearchTree
- root : TreeNode *
constructor
destructor
+ BinarySearchTree ():
+ ~BinarySearchTree ():
+ addToTree (key : int)
+ removeNode (key : int) : bool
+ getHeightOfTree () : int
+ getNumberOfTreeNodes () : int
+ contains (key : int) : bool
+ getRoot () : TreeNode *
+ isEmpty () : bool
+ clear ()
+ printNodeFromTree (node : TreeNode *)
+ printInOrder ()
+ printReverseOrder ()
+ printPreOrder ()
+ printPostOrder ()
+ printDepthFirst ()
+ printBreadthFirst ()
+ printRange (low : int , high : int)
- deleteTree (node : TreeNode *)
- getHeight (node : TreeNode *)
- printInOrderHelper (node : TreeNode *)
- printReverseOrderHelper (node : TreeNode *)
- printPreOrderHelper (node : TreeNode *)
- printPostOrderHelper (node : TreeNode *)
- printRangeHelper (node : TreeNode * , low : int , high : int)
0..*
1
TreeNode
+ key : int
+ numberOfNodes : int
+ height : int
+ left : TreeNode *
+ right : TreeNode *
+ fifoNode : DllNode *
constructor
+ TreeNode (k : int = 0 , numNodes : int = 1 , h : int = 0 , l : TreeNode * = nullptr
r : TreeNode * = nullptr)

Sequence Diagram


cacheManager : CacheManager
hashTable : HashTable
doublyLinkedList : DoublyLinkedList
binarySearchTree : BinarySearchTree
dllNode : DllNode
hashNode : HashNode
treeNode : TreeNode
<<creates>>
<<creates>>
<<creates>>
<<create>>
<< create>>
getTable()
hashTable
getList()
doublyLinkedList
getBst()
binarySearchTree
getSize()
getNumberOfItems()
numberOfItems
isEmpty()
getSize()
add(currKey, myNode)
add(currKey, newNode)
getSize()
insertAtHead(currKey, myNode)
isEmpty()
OK
remove(doublyLinkedList->tail->key)
calculateHashCode(currKey)
true
add(currKey, newNode)
true
removeTailNode()
isEmpty()
true
insertAtHead(currKey, myNode)
isEmpty()
true
insert(node, key)
insert(node, key)
addToTree(key, dllNode)
OK
removeNode(doublyLinkedList->tail->key)
contains(key)
deletion(node, key)
getSuccessor(curr)
deletion(node, key)
getSuccessor(curr)
true
remove(currKey)
contains(currKey)
remove(currKey)
isEmpty()
OK
remove(currKey)
calculateHashCode(currKey)
true
removeNode(currKey)
contains(key)
deletion(node, key)
getSuccessor(curr)
deletion(node, key)
getSuccessor(curr)
true
clear()
clear()
OK
clear()
OK
clear()
deleteTree(node)
deleteTree(node)
OK
getItem(currKey)
getItem(currKey)
contains(currKey)
Item
calculateHashCode(currKey)
getMaxCacheSize()
maxCacheSize
contains(currKey)
contains(currKey)
getItem(currKey)
true
printCache()
printList()
OK
printTable()
OK
printInOrder()
printInOrderHelper(node)
printInOrderHelper(node)
OK
printReverseOrder()
printReverseOrderHelper(node)
printReverseOrderHelper(node)
OK